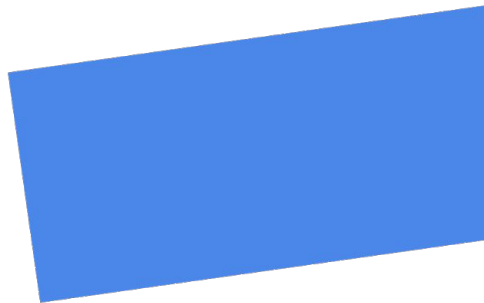
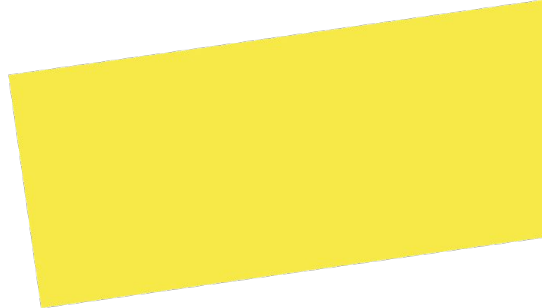




HTML



What is HTML?

HTML = HyperText Markup Language

Describes the structure of web pages

Uses tags to mark up content

Rendered by browsers (Chrome, Edge,
Firefox, Safari)

Brief History of HTML

1991: Tim Berners-Lee proposes HTML

1995: HTML 2.0 standardized (IETF)

1997–1999: HTML 3.2 and 4.01 (W3C)

2000s: XHTML attempts XML-strict HTML

2014: HTML5 recommendation; living standard (WHATWG)

Today: Ongoing updates (semantic tags, forms, multimedia)

Basic Page Skeleton

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width,
initial-scale=1.0" />
    <title>My First Page</title>
  </head>
  <body>
    <h1>Hello, world!</h1>
    <p>This is my first HTML page.</p>
  </body>
</html>
```

Basic Tags You'll Use Often

Headings

`<h1>` to `<h6>`: page and section titles

Paragraphs

`<p>` blocks of text

Links

`<a href="...">Visit</a>`

Images

`` (alt is essential for accessibility)

Lists

`<ul>/<ol>` with `<li>` items

Div/Span

Generic containers for layout (`<div>`) or inline (`<span>`)

Semantic Tags

`<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<footer>`

Meta Tags

Placed inside the <head> section of an HTML document

Common uses

<meta charset="UTF-8"> → character encoding

<meta name="viewport" content="width=device-width, initial-scale=1.0"> → responsive design

<meta name="description" content="Short description of the page"> → SEO

<meta name="keywords" content="HTML, tutorial"> → (less used today)

<meta http-equiv="refresh" content="30"> → auto refresh after 30s

Meta Tag Example

```
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width,  
initial-scale=1.0">  
  <meta name="description" content="Basic HTML tutorial">  
  <title>Meta Example</title>  
</head>
```

Common Tags

```
<h2>About Me</h2>
<p>I like building the web.</p>
<a href="https://example.com">My Portfolio</a>

<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```


Forms & the <input> Tag

<form>

action (where data goes), method (GET/POST)

<label>

Connects text to inputs via for/id; improves accessibility

<input>

type attribute defines behavior (text, email, date, etc.)

Validation

required, minlength, maxlength, pattern

Grouping

<fieldset> and <legend> for related controls

HTML5 <input>

Type	What it does
text	Free-form single-line text
email	Email address with basic validation & mobile @ keyboard
url	Website URL validation
password	Masked characters
number	Numeric input with spinner; use min/max/step
tel	Telephone input; mobile shows dialpad
search	Search field semantics
date	Date picker (browser support varies)
time	Time picker
datetime-local	Local date & time (no timezone)
month	Month picker
week	Week picker

HTML5 <input>

Type	What it does
color	Color picker
range	Slider control; use min/max/step
checkbox	Boolean on/off; multiple values
radio	Single selection within named group
file	File upload
hidden	Not shown; submits value
submit/reset/button	Form buttons

<form> with Labels & Validation

```
<form action="/subscribe" method="post">  
  <label for="email">Email:</label>  
  <input id="email" name="email" type="email" required />
```

```
  <label for="age">Age:</label>  
  <input id="age" name="age" type="number" min="1" max="120" />
```

```
  <label for="fav">Favorite Color:</label>  
  <input id="fav" name="fav" type="color" />
```

```
  <button type="submit">Subscribe</button>  
</form>
```

Tables: When & How

Use tables for data, not layout

Core elements

`<table>`, `<tr>`, `<th>`, `<td>`

Optional structure

`<thead>`, `<tbody>`, `<tfoot>`, `<caption>`

Accessibility

Use `<th scope>` and captions; avoid empty cells

Styling with CSS

Borders, padding, zebra stripes improve readability

Table Example

```
<table>
  <caption>Monthly Sales</caption>
  <thead>
    <tr><th scope="col">Month</th><th scope="col">Amount</th></tr>
  </thead>
  <tbody>
    <tr><th scope="row">January</th><td>$12,000</td></tr>
    <tr><th scope="row">February</th><td>$15,400</td></tr>
  </tbody>
  <tfoot>
    <tr><th scope="row">Total</th><td>$27,400</td></tr>
  </tfoot>
</table>
```

Select and Textarea Tags

<select>

Drop-down menu for multiple choices

Use <option> for items

Can use 'multiple' attribute to allow multiple selections

<textarea>

Multi-line text input field

Attributes: rows, cols, placeholder

Resizable by default in most browsers

Form with <select> and <textarea>

```
<form>
  <label for="country">Choose country:</label>
  <select id="country" name="country">
    <option value="in">India</option>
    <option value="us">USA</option>
    <option value="uk">UK</option>
  </select>

  <label for="feedback">Feedback:</label>
  <textarea id="feedback" name="feedback" rows="4" cols="40">
</textarea>

  <button type="submit">Submit</button>
</form>
```